

Usability and Accessibility Test Report of VotingWorks VxScan, Version 4 with 39 participants for VVSG 2.0, Section 8.3

Report Based on ISO/IEC 25062:2006 Common Industry Format For Usability Test Reports

**VotingWorks VxSuite, Version 4, focused on VxScan with audio
and PAT jacks.**

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1.0 Executive Summary

A usability test of VxScan was conducted between April 15 and May 4, 2026 at various locations in the San Francisco Bay area by VotingWorks. The purpose of this test was to fulfill Section 8.3 of the Voluntary Voting System Guidelines (VVSG).

During the usability tests, a total of 39 testers used the VxScan in a simulated election. The election consisted of one test ballot with 12 contests, including:

- State and Local contests
- Partisan and nonpartisan contests
- Single member and multimember contests
- Ballot questions (retention races)
- Ballot propositions (reflecting referendum, ballot initiatives, and constitutional amendments)

This ballot used for the test is based on the test ballot developed by the National Institute of Standards and Technology (NIST), to provide opportunities for various voting tasks. The instructions includes 22 tasks and that model typical ballots from around the country, including:

- Voting for names at various locations within a list of names
- Voting a partial slate in a multimember contest
- Skipping elements of a ballot
- Changing a vote from the review screen
- Write-in votes
- Reviewing a ballot
- Casting a ballot

Following the conclusion of the testing, the results were analyzed to determine participants' effectiveness, efficiency and satisfaction using the VxScan.

During the usability test, the testing team collected and analyzed the following types of data:

- Number and percentage of ballots successfully submitted/completed (number and percentage)

- Percent of tasks completed without any errors
- Count of assists provided (number or percent of testers needing assists)
- Skipped contests
- Time to complete the voting session
- Rating of Audio
- Rating of Translations
- testers' confidence that they had used the system correctly
- testers' overall satisfaction with the system

The testing team also collected less structured feedback on the VxMark during, and after, voting sessions, which included general feedback on the voter's experience during the session and recommendations from the voter on improvements to the system.

Below is a high-level summary of the results:

Measure	Description	Usability Test Results
Successful completion	Average success rate of the 22 tasks performed by testers from the general population.	100.0%
Ballots cast without any assists	Count of the number of testers who were able to submit their ballot without any errors requiring assistance.	37 of 39 ballots (95.0%)
Count of assists provided	Number of testers and assists provided to testers during the usability test.	5 Assists given to 3 testers
VotingSession Time	Mean time taken per test participant to complete the process of activating, filing out and casting the ballot. (Note: Voters were	15 m 45 s

using the VxMark to mark a ballot, so this time is calculated based on the entire voting experience. The VxMark is not part of this assessment at this time. The average time a tester took to cast their ballot on VxScan was less than 30 seconds.)

Rating of Audio quality	Number of audio users and percent who thought audio worked well	25 of 39 testers used this 64.1% thought the audio worked well
Rating of Translations	Number of audio users and percent who thought translations worked well	6 of 39 testers used this 100.0% thought the translations worked well
Voter Confidence	Mean confidence level expressed by testers that they believed they voted correctly and the system successfully recorded their votes.	4.44 Confidence Level
Voter Satisfaction	Mean satisfaction level expressed by testers in response to a post-test satisfaction questionnaire focused on VxScan.	4.04 Satisfaction Level

2.0 Introduction

2.1 Full Product Description

During the usability test the VxScan version 4 was evaluated. The VxMark version 4 was used for ballot marking and review, but is not included in the results of this report, as it is not part of the VxSuite 4.0 campaign. This version is the same as what was provided to the Voting System Test Laboratory with SC-15 versions of the scanner.

The VxScan is a precinct tabulator that is used in the polling place for voters to cast handmarked paper ballots. In future releases, the VxScan will be able to accept machine marked paper ballots, but that is not part of the current VxSuite 4.0 certification process. The focus of VxScan testing was on the voter experience casting the ballot and using the audio and PAT input to respond to system notifications. VxScan does not have a ballot review screen, so this review focuses solely on the activity of casting the ballot on VxScan.

2.2 Test Objectives

The usability test objectives include:

- To assess the effectiveness of the VxScan by measuring the abilities of users to successfully cast a ballot.
- To assess the efficiency of the product by measuring the average time to complete a voting session. This time calculation is based on the full voting session using VxMark to mark and verify the ballot and VxScan to cast the ballot. It was too confusing to the testers to just track the time taken to cast the ballot, because the average time was less than 30 seconds to complete the action of casting a ballot.
- To assess the user satisfaction of the system by measuring average voter confidence.
- To gather feedback on the experience of using audio and translations.
- To gather open ended, unstructured feedback on the system and voter experience.

3.0 Method

3.1 Participants

Participants were recruited by VotingWorks staff via in person outreach.

A total of 39 testers, with a varying mix of backgrounds and demographic characteristics were selected to participate in the usability test. All participants were over the age of 18, eligible to vote in the U.S., and able to communicate in English.

The usability tests used individuals whose visual acuity is less than 20/70 but greater than or equal to 20/200; the usability tests used individuals who can only read large-print, high contrast text; and used individuals who are visually impaired. Wearing corrective lenses did not assist these individuals in reading normal-sized text. VotingWorks had 12 testers with low vision who were able to complete the testing session, without assistance. More information available in the table below and Appendix A.

Please see Appendix A for a full spreadsheet of participant demographics.

Breakdown by demographics for 71 total participants

Demographic	All Participants
Sex	[38] Men
	[26] Women
	[1] Other

Age	[0] 18-24 years
	[7] 25-34 years
	[12] 35-44 years
	[8] 45-54 years
	[6] 55-69 years
	[6] 70+ years
Education	[6] Some high school
	[3] High school
	[11] Some college
	[16] College degree
	[3] Graduate/professional degree
Race/Ethnicity	[1] American Indian/Alaska Native
	[6] Asian
	[5] Black/African-American
	[10] Hispanic/Latino
	[1] Pacific Islander
	[16] White/Caucasian

Disability	[14] Blind
*Many testers identified more than one disability	[10] Low vision
	[1] Deaf
	[1] Hard of Hearing
	[9] Limited dexterity
	[1] Intellectual and Developmental disability
	[0] Limited or low literacy
	[2] Mobility
	[0] Brain/cognitive
	[6] NA
Language Used	[36] English
*several testers noted they were bilingual and chose to vote a ballot in another language	[5] Spanish
	[1] Chinese (Traditional)
	[1] Chinese (Simplified)
Geographic Distribution	[19] Urban
	[19] Suburban
	[1] Rural

Years of Voting Experience	[7] None/Chose not to answer
	[2] 2-5 years
	[3] 6-10 years
	[8] 11-20 years
	[19] 20+ years

Number of elections in the past 4 years	[8] None
	[6] 1-2
	[7] 3-5
	[18] 6+

3.2 Context of Use in the Test

3.2.1 Tasks

During the usability test, participants were instructed to vote in a simulated election consisting of one test ballot with 12 contests, including:

- State and Local contests
- Partisan and nonpartisan contests
- Single member and multimember contests
- Ballot questions (retention races)
- Ballot propositions (reflecting referendum, ballot initiatives, and constitutional amendments)

Our usability testing team based the test ballot on the medium complexity ballot created by NIST in order to cover all of the possible errors it incorporates.

Using this ballot, participants were asked to perform 22 tasks that were selected to model typical ballots from around the country, as well as to thoroughly test the voting system's capabilities and usability, including:

- Voting for names at various locations within a list of names
- Voting a partial slate in a multimember contest

- Skipping elements of a ballot
- Changing a vote from the review screen
- Write-in votes
- Reviewing a ballot
- Casting a ballot

Participants were provided an overview of the system, description of all tools and accessible options, and were asked to perform the task of marking, reviewing and casting a ballot without assistance. A task was considered successful if the participant was able to cast a vote in a way that matched their preference.

Data was collected for each task, including successful completions, time to complete a voting session, number of errors requiring assistance, number of assists provided, and voter confidence for each task.

It is important to note that although testers completed all the tasks listed above, this report focuses on the experience of casting a ballot on VxScan. All information regarding marking and verification of the ballot is excluded from this report.

Please see Appendix D for the test ballot.

3.2.2 Test Location

The VxScan is intended to be used at designated polling locations across the U.S., including schools, libraries, churches and other public facilities large enough to house multiple voting stations.

In order to simulate this environment, the test was conducted at several locations:

- Private residences for language testers
- Ability Now, Bay Area
- SF Lighthouse

3.2.3 Voting Environment

During an actual election, testers are expected to use the voting system provided at the polling location. Testers may have experience with a wide-range of systems or may only have experience with one type of system.

During the usability test, all participants were instructed to use VxScan as if this system was implemented at their local polling location. They were provided an

overview of the equipment, accessible features, and then told to begin their voting session. The test administrator only intervened if assistance was requested by the voter.

3.2.3.1 Display Devices

The VxScan uses a 15.6" screen, with resolution of 1920x1080, and 310 nit brightness.

3.2.3.2 Audio Devices

VxScan makes noises whenever a ballot is accepted or rejected. The ballot accepted sound is a pleasant chime and the ballot rejected sound is a jarring beep. Election managers can toggle sounds on or off. When no headphones are plugged into the front panel headphone jack, the sound comes out of the onboard speaker. When headphones are plugged in for voter accessibility, audio output is routed to the headphones via the single-board computer's audio output.

3.2.3.3 Input Devices

During the test, participants could use: headphones, jelly buttons, and/or the touch screen. We also offered a stylus, if that made it easier to use the touchscreen.

3.2.4 Test Administrator Tools

During the usability test, various tools were used to facilitate the test sessions, including:

- Informed Consent (See Appendix B)
- Instructions for Participants (See Appendix E)
- Post-test Satisfaction Questionnaire (See Appendix C)

Participants' votes were recorded by the system, similar to a real-world election. Test facilitators recorded time by entering that information on the questionnaire. Start time (when the tester clicked the screen to bring up their ballot) and end time (when the voter cast their ballot) are recorded in the questionnaire and results.

3.3 Methodology

During the usability test, participants interacted VxScan. Each participant used the system set up in the same way and was provided with the same ballot, system orientation, and instructions.

The system was evaluated for effectiveness, efficiency and satisfaction. To evaluate these factors, the usability team collected data on:

- Number and percentage of ballots successfully submitted/completed number and percentage
- Percent of tasks completed without any errors
- Count of assists provided (number or percent of testers needing assists)
- Skipped contests
- Time to complete the voting session
- Rating of Audio
- Rating of Translations
- Testers' confidence that they had used VxScan correctly
- Testers' overall satisfaction with VxScan

Additional information about the various measures and associated metrics can be found in the section on Usability Metrics.

3.3.1 Procedure

Upon arrival, participants were greeted and asked to complete a Pre-Test Questionnaire (See Appendix C) to ensure that they qualified for the test. Participants who did not meet the qualifications were allowed to test and provide feedback, but their results are excluded from the data analyzed in this report.

Participants meeting the qualifications were asked to review and sign an Informed Consent (See Appendix B), which described their rights during the study and was provided in four languages: English, Spanish, Chinese (Traditional), and Chinese (Simplified). Participants were then provided an overview of the voting system and assistive technology options. Test administrators described the different settings that could be used by the tester and helped orient anyone using the accessible controller, so they could feel the buttons and easily identify the help button on the controller.

After this overview, they were given the following instructions:

"Read this to tester right before you begin the test: "Please attempt to follow the instructions provided on the following screens and/or through audio provided. I will be here to provide basic assistance, similar to what a poll worker can provide in a polling location. Please feel free to share your feedback and comments out loud as you work through the ballot. If you are stuck and cannot continue, please inform me

by raising your hand and/or verbally asking for help. I will be observing and taking notes from a few feet away.”

During the usability test, test facilitators observed users’ interactions and noted feedback they provided as they worked through the ballot. The test administrator only provided assistance when a user was stuck and asked a question. The assistance was limited to the question asked. Test administrators did not walk the voter through the ballot. Test administrators noted the start and end time of each session. When the participants finished the test, they were asked to complete a Post-Test Questionnaire (See Appendix C).

At the conclusion of the test, participants were thanked for their time and compensated with gifts cards for their time and energy.

Two staff members participated in this test.

3.3.2 Participant General Instructions

The participants were instructed that they would work alone during the test, but that they could ask for the same sort of assistance or the same sort of questions they might ask an election worker.

3.3.3 Participant Task Instructions

Participants were also provided with instructions on how to use different accessible features, reminded to read instructions and follow screen prompts, and to make sure they reviewed their selections on screen prior to printing their ballot. They were reminded and encouraged to write in candidates and review and change selections as they worked through the ballot. These instructions were provided to users orally.

See Appendix E for the instructions for participants.

3.4 Usability Metrics

The usability test collected various metrics for effectiveness, efficiency and satisfaction.

Name	Measure	Description
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Effectiveness: Completion Rate	Ballots successfully submitted/completed	Percentage of test participants who were able to complete the process of voting and cast their ballots so that their ballot choices were recorded by the system.
Effectiveness: Errors	Tasks completed according to the on screen instructions without errors or assistance	Percentage of tasks that were completed without any errors. An error might involve a voter not following screen instructions to navigate through the ballot
Effectiveness: Assists	Number of participants needing assists	Count and percentage of the number of participants needing assists
Effectiveness: Assists	Count of assists provided	Count of the number of times assistance was given to participants.
Efficiency: Time on task	Average session time	Mean time taken per test participant to complete the process of activating, marking, and casting the ballot, both overall and by interaction mode
Satisfaction: Audio	Satisfaction with audio quality	Satisfaction expressed by testers in response to a post-test satisfaction questionnaire.

Satisfaction: Translations	Satisfaction with translation quality	Satisfaction expressed by testers in response to a post-test satisfaction questionnaire.
Satisfaction: Overall Satisfaction Rating	Average voter satisfaction	Mean satisfaction level expressed by testers in response to a post-test satisfaction questionnaire.
Satisfaction: Confidence	Average voter confidence	Mean confidence level expressed by testers that they believed they voted correctly and the system successfully recorded their votes.

4.0 Results

4.1 Data Analysis

To analyze the data, each session was scored for completeness, accuracy, errors, number of assists, and time to complete. In addition, the test team analyzed testers' satisfaction and confidence using various post-test questionnaires.

4.2 Presentation of the Results

This section details the performance results for effectiveness (completion rate, errors, assists), efficiency (time on task) and satisfaction (satisfaction and confidence rating).

Name	Measure	Results
Effectiveness: Completion Rate	Ballots successfully submitted/completed	Of the 39 participants included in the test, 39 were able to successfully cast their ballot.

Effectiveness: Errors	Tasks completed according to the instructions without any errors	39 of 39 (100.0%) testers were able to complete a voting session according to instructions without any assists
Effectiveness: Assists	Number of participants needing assists	2 [5.0%] of the participants needed assists
Effectiveness: Assists	Count of assists provided	A total of 5 assists were provided to those participants.
Efficiency: Time on task	Average session time	The average session time was 15 minutes 45 seconds. (Note: This session time includes the time taken to mark and verify a ballot prior to casting. This report focuses on the activity of casting on VxScan alone .)
Satisfaction: Audio	Satisfaction with audio quality	25 of 39 testers used audio during their session. All said it was satisfactory on VxScan. (Asked as a YES/NO question)
Satisfaction: Translations	Satisfaction with translation quality	6 of 39 testers used translations. All 6 testers thought the translations were satisfactory. (Asked as a YES/NO question)
Satisfaction: Overall	Average voter satisfaction	Average voter satisfaction using vxScan to cast a ballot was 4.04.

Satisfaction Rating		Testers were asked to give their overall satisfaction rating on a scale of 1-5, with 1 being poor and 5 being excellent.
Satisfaction: Confidence	Average voter confidence	Average voter confidence was 4.44. Testers were asked to give their confidence rating on a scale of 1-5, with 1 being poor and 5 being excellent.

More detailed results can be found in Appendix F.

5.0 Conclusion

Overall

Most testers expressed that they had a very positive experience and the test to cast on VxScan was the shortest part of the session, less than a minute for all testers.

There wasn't a lot of VxScan specific feedback, so here is a quick summary:

- One voter couldn't use the scanner without assistance, because they had limited dexterity and feeding the paper into the slot was difficult.
- One voter stated that the ballot insert tray was low contrast and a bit hard to see.
- One voter felt that the scanner height was a little high.
- One voter had a little trouble inserting the ballot into the slot, but appreciated the audio that indicated what was happening.
- One noted that the scanner announced their ballot was counted in English and then had a conversation with the tester about language settings.

Accidental Activation

Test administrators did not witness or have a tester report to them an instance of accidental activation of actions through an unintended gesture via the touchscreen or other accessible inputs.

Appendixes

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- [Appendix B: Informed Consent](#)
- [Appendix C: Test Session Questionnaire](#)
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Appendix A: Participant Demographics

 Copy of UAT Data Collection Form 2026 (Responses)

Appendix B: Informed Consent

 [Copy of Informed Consent Form - VxMark_VxScan UAT SF - 2026 - Chinese, sim...](#)

 [Copy of Informed Consent Form - VxMark_VxScan UAT SF - 2026 - Spanish.doc...](#)

 [Copy of Informed Consent Form - VxMark_VxScan UAT SF - 2026 - English.docx...](#)

 [Copy of Informed Consent Form - VxMark_VxScan UAT SF - 2026 - Chinese, tra...](#)

Appendix C:Test Session Questionnaire

 Copy of UAT Data Collection Form 2026 - Google Forms.pdf

Appendix D: Test Ballot Specification

 PROOF-official-precinct-ballot-Precinct_1-1_zh-Hant.pdf

 PROOF-official-precinct-ballot-Precinct_1-1_zh-Hans.pdf

 PROOF-official-precinct-ballot-Precinct_1-1_es-US.pdf

 PROOF-official-precinct-ballot-Precinct_1-1_en.pdf

Appendix E: Instructions for Participants

Prior to starting the session, test participants were given the following instructions orally after receiving an overview of the accessible options of the VxMark & VxScan:

1. Please let me know if you need headphones and/or tactile input.
2. Please use the settings menu to select the contrast , language, and font size that works best for you.
3. Please read the instructions at the beginning of each contest carefully and pay attention to screen alerts. Work through each contest selecting the candidates you choose.
4. Please complete a write in entry for a contest of your choosing or make changes in any contest or on the review screen prior to printing your ballot.
5. Please review all your selections after you've completed your ballot and make any corrections you believe are necessary.
6. Please press the cast ballot option to print your ballot.
7. After printing your ballot, we will proceed to the precinct scanner for you to cast your ballot. Please let me know when you are ready to cast your ballot.
 - a. At the scanner - Please let me know if you need headphones and/or tactile input for the precinct scanner. I will remind you of the settings options for VxScan when you are ready to cast your ballot.
8. The test administrator is available to answer questions as a poll worker, but will not otherwise assist you with your test session.
9. Please feel free to speak feedback or frustrations out loud, the test administrator will note that feedback for system improvements.
1. When you are ready, please select the button at the bottom of the screen. n.

Appendix F: Results

The results are available here:

 [Copy of UAT Data Collection Form 2026 \(Responses\)](#)